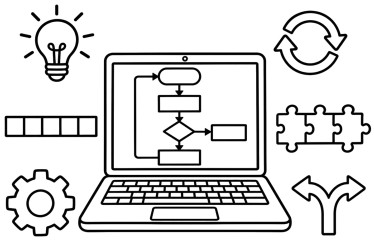
# TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

# UNIT 1 - PARTS 1 AND 2

## Programming



**(!) FOR THE TEACHER ONLY**

This booklet has the answers to every exercise of the two Unit 1 Student Workbooks (Part 1 and Part 2), in the same order. The exercise letters match the workbooks.

**(i) WHAT IS IN THIS GUIDE**

- **Marks and pacing** - both workbooks, and one class marks record
- **Practicals at a glance** - Practicals 1 - 6, the software, a quick quiz
- **Writing spaces** - what students write in every writing space
- **Answers** - Part 1 (chapters 1 - 3), then Part 2 (chapters 4 - 6)
- **Two tests** - the marking schemes of the Part 1 and Part 2 tests
- **Extra** - common mistakes, textbook misprints, homework, tips

**(★) BEFORE THE FIRST LESSON**

- ☐ Free Pascal (or Lazarus) runs on every lab computer before Practical 3.
- ☐ Every student has both workbooks: Part 1 (1.1 - 1.5) and Part 2 (1.6 - 1.11).
- ☐ You have the notes of the writing spaces ready (in this guide).

"Let them type, run and fix real programs. Programmers are made at the keyboard."

- THE KING OF ICT

TEACHER GUIDE

# Marks and Pacing

Both workbooks: Part 1 (chapters 1 - 3) and Part 2 (chapters 4 - 6)

| Part / chapter | Exercise | Marks |
|---|---|---|
| **Part 1 - Ch 1** | A Input, process and output | 9 |
| | B Sequence, selection or repetition? | 8 |
| | C Put the algorithm in order | 8 |
| | D True or false | 6 |
| **Part 1 - Ch 2** | A Name the symbols | 6 |
| | B Complete the flowchart | 7 |
| | C Convert the flowchart to pseudo code | 6 |
| | D Follow the pseudo code | 6 |
| **Part 1 - Ch 3** | A Valid or not valid? | 8 |
| | B Which data type? | 8 |
| | C Work out the results | 8 |
| | D Find the reserved words | 6 |
| | E Find the errors | 5 |
| **Part 1** | **Unit 1 Part 1 Test** | **40** |

| Part / chapter | Exercise | Marks |
|---|---|---|
| **Part 2 - Ch 4** | A What is the output? | 8 |
| | B Trace the loops | 6 |
| | C Complete the program | 6 |
| | D Find the errors | 6 |
| **Part 2 - Ch 5** | A Fill the array | 7 |
| | B What does orderNos print? | 6 |
| | C What does it print? | 6 |
| | D Complete the program | 5 |
| **Part 2 - Ch 6** | A Procedure or function? | 6 |
| | B Which language level? | 8 |
| | C Assembler, interpreter or compiler? | 6 |
| **Part 2** | **Unit 1 Part 2 Test** | **40** |
| **TOTAL** | | **241** |

| Part | Chapters | Textbook pages | Suggested periods |
|---|---|---|---|
| Part 1 | 1 Problem solving and algorithms | 1 - 9 | 4 |
| | 2 Flowcharts and pseudo codes | 9 - 17 | 5 |
| | 3 Pascal basics and operators | 17 - 25 | 6 |
| Part 2 | 4 Selection and repetition | 25 - 33 | 7 |
| | 5 Nested structures and arrays | 33 - 38 | 5 |
| | 6 Sub-programs and programming languages | 38 - 52 | 6 |

## ! SUGGESTED SHAPE OF ONE CHAPTER

| | |
|---|---|
| **READ** | Teach the chapter pages with the textbook open. Students underline the key words. |
| **WRITE** | Give the notes of the WRITING SPACES (this guide lists them all). Students write them neatly. |
| **DO** | The practical: paper activities in Part 1 chapters 1 - 2, Free Pascal from chapter 3 on. |
| **CHECK** | The written exercises, then mark them together. Each part ends with its own 40-mark test. |

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# Class Marks Record

Copy this page for each class - both parts on one page

| No | Student name | Ch 1 (31) | Ch 2 (25) | Ch 3 (35) | Test 1 (40) | Ch 4 (26) | Ch 5 (24) | Ch 6 (20) | Test 2 (40) | Total (241) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
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| 4 | | | | | | | | | | |
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| 28 | | | | | | | | | | |
| 29 | | | | | | | | | | |

TEACHER GUIDE

# Practicals at a Glance

The textbook Activities and the Pascal programs

| Practical | Students do | Check that ... |
|---|---|---|
| **1 (Activities pp. 3 - 9)** | input, process and output of two problems; an algorithm for the three houses; incidents of sequence, selection and repetition; the voting selection | toffees: input 100 toffees and 20 people, process 100 / 20, output 5 toffees each; the vote: more, can, cannot. |
| **2 (Activity p. 12)** | flowcharts for five problems; two of them as pseudo codes | the symbols of Table 1.1; one Start and one End; a decision has Yes and No arrows. |
| **3 (pp. 24 - 25)** | addNum in Free Pascal: compile, run, read an error message | 12 and 8 give Total is 20; the missing ; gives a compile error before the program runs. |
| **4 (pp. 26 - 33)** | positiveNo, GradeForMarks, FindGrade, print10Nos, printReverse, total_avg, the REPEAT examples | 80, 68, 52, 40, 20 give A, B, C, S, F; FindGrade with 120 prints Invalid Marks; E.g. 2 prints only 5, E.g. 3 prints 5 to 50. |
| **5 (pp. 34 - 38)** | orderNos with A, D and a; rep_sel with ten numbers; ictMarks for 5 students | a small a prints nothing more; the counts agree with the students' own counts; max and avg are right. |
| **6 (pp. 41 - 42, 49)** | procedure_circle and function_circle; a compile error; the Car class; the paradigms | radius 7 gives 154 and 44; objects such as Nimal's car, properties such as colour, methods such as start. |

## ! SOFTWARE AND MATERIALS

- Free Pascal (freepascal.org) with its text IDE, or Lazarus, on every lab computer. In the IDE: File New, Alt+F9 compile, Ctrl+F9 run, Alt+F5 the output screen.
- In Lazarus, add readln; before end. so that the console window stays open.
- The textbook has no Activity boxes in the Pascal sections (1.4 - 1.10): Practicals 3 - 6 run the textbook's own programs, as printed in the workbooks.

## ? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

| Ask the class ... | Answer |
|---|---|
| 1. Input, process, output: what is converting the input to the output? | **the process** |
| 2. Name the three control structures. | **sequence, selection, repetition** |
| 3. Which flowchart symbol is a decision? | **the diamond** |
| 4. What is 20 DIV 6? And 20 MOD 6? | **3 and 2** |
| 5. Which operator has the highest priority? | **NOT** |
| 6. How many times does FOR X := 5 TO 10 DO repeat? | **6** |
| 7. Which loop checks its condition at the beginning? | **WHILE ... DO** |
| 8. How many elements has array[0..4]? | **5** |
| 9. A sub-program that returns an output is a ... | **function** |
| 10. Which translator translates the whole program first? | **compiler** |

WRITING SPACES

# What Students Write

Read each note out or write it on the board - the students copy it into the space

## Other pages

| No. | Page | What the students write |
|---|---|---|
| 1 | Part 1 p. 2 | **What is a program?** A program is a sequence of instructions which performs a certain task using the computer. A computer language is needed to provide the instructions. |

## Chapter 1

| No. | Page | What the students write |
|---|---|---|
| 2 | Part 1 p. 3 | **Input, process and output** Input: the raw materials used to solve a problem. Output: the result obtained after solving it. Process: converting the input to the output, step by step and in the correct order. |
| 3 | Part 1 p. 3 | **Alternative solutions** More than one solution to a problem are alternative solutions; all of them are the solution space. We select the most appropriate one, so that the program is short and simple. |
| 4 | Part 1 p. 5 | **Three control structures** Sequence: steps carried out in a strict order. Selection: steps executed depending on whether a condition is satisfied. Repetition: steps repeated until a condition is satisfied. |
| 5 | Part 1 p. 5 | **Examples of selection** Passing a subject: if the mark is 35 or more it is a Pass, if not a Fail. Buying a book: if you have money equal to or more than its price you can buy it, if not you cannot. |

## Chapter 2

| No. | Page | What the students write |
|---|---|---|
| 6 | Part 1 p. 8 | **Flowchart symbols** Rounded box: start or end. Parallelogram: input or output. Rectangle: process. Diamond: decision. Arrow: flow direction. Small circle: connector. |
| 7 | Part 1 p. 9 | **Pseudo code** A pseudo code presents an algorithm in simple English terms. It is independent of a computer language, so it can be converted to the instructions of any programming language. |

### TEACHING TIPS

- Start Part 1 with a real problem from school life: the bus broke down. Let the class list every way to reach school - that list is the solution space.
- For every problem, draw three columns on the board: INPUT, PROCESS, OUTPUT. Students fill them before any flowchart or program.
- Write ten identifiers on the board; the class shouts valid or not valid and gives the rule.

## Chapter 3

| No. | Page | What the students write |
|---|---|---|
| 8 | Part 1 p. 12 | **Identifiers** An identifier names a variable, a constant or a program. It starts with an English letter; then letters, numbers or _. No spaces, no special characters, no reserved words. |
| 9 | Part 1 p. 14 | **Operator precedence** 1. NOT 2. * / DIV MOD AND 3. + - OR 4. = < > < <= > >=. An operator of a higher priority is evaluated first; brackets are evaluated first of all. |
| 10 | Part 1 p. 16 | **Data types** Integer: whole numbers (46, -12). Real: decimal numbers (25.68). Boolean: True or False. Char: one character ('k'). String: a sequence of characters ('ICT'). |

## Other pages

| No. | Page | What the students write |
|---|---|---|
| 11 | Part 2 p. 2 | **Pascal** Pascal is a high-level programming language. It has the features of the procedural paradigm and of structured programming. |

## Chapter 4

| No. | Page | What the students write |
|---|---|---|
| 12 | Part 2 p. 4 | **IF statements** IF ... THEN: the statement runs only if the condition is satisfied. IF ... THEN ... ELSE: statement 1 if it is satisfied, if not statement 2. Nested IF: many conditions. |
| 13 | Part 2 p. 6 | **FOR, WHILE and REPEAT** FOR ... DO: the number of repetitions is known in advance. WHILE ... DO: the condition is checked at the beginning. REPEAT ... UNTIL: checked after one run; stops when true. |
| 14 | Part 2 p. 6 | **The CASE statement** When one variable has multiple conditions, a CASE statement is easier than repeated IF ... THEN ... ELSE statements: Case Marks of 0..34 : ... Else ... End; |

## Chapter 5

| No. | Page | What the students write |
|---|---|---|
| 15 | Part 2 p. 8 | **Nested control structures** A control structure may be included inside another one: a repetition inside a selection (orderNos) or a selection inside a repetition (rep_sel). |
| 16 | Part 2 p. 9 | **Arrays** An array saves data items of the same type in adjoining memory spaces under a single variable name. Each element has an index in square brackets: num[0], num[1] ... |

### TEACHING TIPS

- Ask the class to act out an algorithm: one student reads the steps, another follows them exactly.
- Teach the precedence with a ladder: NOT at the top, the comparisons at the bottom.

## Chapter 6

| No. | Page | What the students write |
|---|---|---|
| 17 | Part 2 p. 12 | **Sub-programs** Sub-programs make a complex program easy to read, understand and maintain. A function returns an output to the main program; a procedure does not. |
| 18 | Part 2 p. 14 | **Low-level and high-level languages** Machine language (0s and 1s) runs directly and fast but depends on the machine. Assembly uses symbols (assembler). High-level languages use English words. |
| 19 | Part 2 p. 15 | **Language translators** Assembler: assembly to machine language. Interpreter: translates and runs one statement at a time, each time. Compiler: translates the whole program once. |
| 20 | Part 2 p. 15 | **Programming paradigms** Imperative: procedural (C), object oriented (Java), parallel processing (Java). Declarative: logic (Prolog), functional (Lisp), database (SQL). |

### TEACHING TIPS

- Let students swap two steps of an algorithm and ask: does the output change? This shows strict order.
- Draw the six flowchart symbols on cards. Students hold up the right card when you read a step.
- Colour the loop-back arrow of every flowchart. Students see at once where the repetition is.
- Trace pseudo code E.g. 4 with a table on the board: n, Number, Total. Let a student be the computer.
- Say it often: Total = Total + Number is an instruction, not an equation. The old value of Total is lost.
- Remind students: the values of Char and String are in single quotes. "Grade" in double quotes is an error.
- Use long division on the board for DIV and MOD: the quotient is DIV, the remainder is MOD.
- Free Pascal is free (freepascal.org) and Lazarus is its visual IDE. Install it on every lab computer before Practical 3.
- In the Free Pascal IDE: Alt+F9 compiles, Ctrl+F9 runs and Alt+F5 shows the output screen.
- Free Pascal prints real numbers in E notation (1.54...E+002 for 154). Show writeln(area:0:2) to print 154.00.
- Let students make one mistake on purpose (delete a ;) and read the compiler message together.
- Before each program is run, students predict the output on paper. Then they run it and check.
- Nested IF: draw the grade flowchart as a staircase. Each No goes one step down.
- CASE: write the ranges 0..34, 35..49 ... on the board and let students put marks into the right range.
- Count FOR repetitions with fingers: FOR X := 5 TO 10 is 5, 6, 7, 8, 9, 10 - six fingers, not five.

ANSWERS - PART 1

# Chapter 1

Problem Solving and Algorithms

## A Input, process and output

9 marks

| Problem | Input | Process | Output |
|---|---|---|---|
| Problem 3: Dividing 40 page and 80 page books from a parcel between Sanduni and Anupama | The parcel of books | Open the parcel; take a book out; a 40 page book goes to Sanduni and an 80 page book to Anupama; go back to taking a book out till all the books are taken out of the parcel. | Sanduni gets the 40 page books and Anupama gets the 80 page books. |
| Problem 6: Finding the larger number between two numbers | Two numbers | Comparing the two numbers and finding the larger one. | The larger number |
| Problem 7: Finding whether a number is odd or even | A number | Dividing the number by 2: even if the remainder is 0, odd if the remainder is 1. | Whether the number is odd or even |

1 mark for each cell (textbook Problems 3, 6 and 7, pages 2 - 3). Accept the same ideas in the student's words.

## B Sequence, selection or repetition?

8 marks

| | | | |
|---|---|---|---|
| **1. Sequence** | **2. Selection** | **3. Repetition** | **4. Sequence** |
| **5. Selection** | **6. Repetition** | **7. Selection** | **8. Repetition** |

1 mark each (the examples of 1.2.1, pages 7 - 9; the last row is the sugar algorithm of page 6).

## C Put the algorithm in order

8 marks

a = **5** b = **1** c = **6** d = **3** e = **4** f = **2**

**1. Which two steps can be interchanged without changing the output? Why? (2 marks)**

Writing the address (4) and sticking the stamp (5): you can stick the stamp after writing the address or write the address after sticking the stamp.

1 mark for each step in the right place (accept 4 and 5 either way).

### TEACHING TIPS

- Show that REPEAT ... UNTIL runs at least once: E.g. 2 prints 5 even though sum < 50 is already true.
- Ask: what happens if the condition of a WHILE loop never becomes false? Let students find the infinite loop.
- Make a class chart of the paradigms with one language each: C, Java, Prolog, Lisp, SQL.

## D True or false

6 marks

1. **T - When analyzing a problem, the input, process and output are identified separately.**
2. **F - A process takes place step by step, but the order of the steps is never important.**
3. **T - In the solution "Perimeter = length + width + length + width" there is no multiplication.**
4. **F - In an algorithm, the order of every step can be changed without changing the output.**
5. **T - There are three types of control structures: sequence, selection and repetition.**
6. **F - In selection, both choices are carried out one after the other.**

ANSWERS - PART 1

# Chapter 2

Flowcharts and Pseudo Codes

## A Name the symbols

6 marks

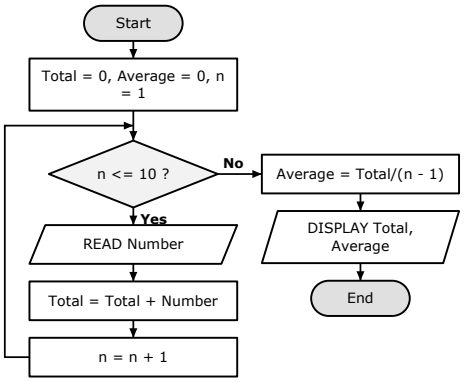
| | | |
|---|---|---|
| **1.** Start or end | **2.** Input or output | **3.** Process |
| **4.** Decision | **5.** Flow direction | **6.** Connector |

1 mark each (Table 1.1, page 10).

## B Complete the flowchart

7 marks

1. **Total = 0, Average = 0, n = 1**
2. **n <= 10 ?**
3. **READ Number**
4. **Total = Total + Number**
5. **n = n + 1**
6. **Average = Total/(n - 1)**
7. **DISPLAY Total, Average**




1 mark each. The flowchart is pseudo code E.g. 4: the loop is WHILE n <= 10 ... ENDWHILE; after the loop comes the average.

### TEACHING TIPS

- Draw an array as a row of boxes on the floor with chalk. Students stand in num[0] ... num[4] holding numbers.
- Let students explain one line of a program in their own words before they type it.

## C Convert the flowchart to pseudo code

6 marks

```
BEGIN
    INPUT Weight
    IF Weight > Standard weight THEN
        Fare = Standard fare + Additional fare
    ELSE
        Fare = Standard fare
    ENDIF
    DISPLAY Fare
END.
```

1 mark each: BEGIN and END; INPUT Weight; IF ... THEN with the condition; the two fares in the right branches; ELSE ... ENDIF; DISPLAY Fare. Accept READ / OUTPUT and the student's own variable names.

## D Follow the pseudo code

6 marks

1. What is the value of Total after the third number? **13**
2. How many times is READ Number executed? **10**
3. What is the value of n when the repetition stops? **11**
4. What is the value of Total at the end? **50**
5. What is the value of Average? **5**
6. Why is Total divided by (n - 1) and not by n? **n is 11 when the loop stops; n - 1 = 10 numbers**

Total after each number: 4, 11, 13, 22, 27, 33, 36, 44, 45, 50. 1 mark each.

ANSWERS - PART 1

# Chapter 3

*Pascal Basics and Operators*

## A Valid or not valid?

8 marks

| | | | |
|---|---|---|---|
| **1.** Student_Name: **valid** | **2.** 2ndTerm: **not valid** | **3.** Total Marks: **not valid** | **4.** Num1: **valid** |
| **5.** avg-mark: **not valid** | **6.** begin: **not valid** | **7.** FirstName: **valid** | **8.** Marks$: **not valid** |

1 mark each. Not valid: 2ndTerm - starts with a number; Total Marks - has a space; avg-mark - has a special character (-); begin - a reserved word; Marks$ - has a special character ($).

## B Which data type?

8 marks

| | | | |
|---|---|---|---|
| **1.** 46: **Integer** | **2.** 25.68: **Real** | **3.** 'k': **Char** | **4.** 'Sri Lanka': **String** |
| **5.** True: **Boolean** | **6.** -12: **Integer** | **7.** '7': **Char** | **8.** 0.0: **Real** |

1 mark each. '7' in quotes is a Char; 0.0 is a Real (page 19).

## C Work out the results

8 marks

| | | | |
|---|---|---|---|
| **1.** 17 DIV 5 = **3** | **2.** 17 MOD 5 = **2** | **3.** 15 / 4 = **3.75** | **4.** 4 + 10 MOD 3 = **5** |
| **5.** 2 + 3 * 4 = **14** | **6.** 18 / 3 * 2 = **12** | **7.** 9 > 4 = **True** | **8.** 6 < > 6 = **False** |

1 mark each. 4 + 10 MOD 3: MOD first (10 MOD 3 = 1), then 4 + 1. 18 / 3 * 2: the same priority, from the left: 6 * 2. 15 / 4 gives a real number.

## D Find the reserved words

6 marks

| | | | | | |
|---|---|---|---|---|---|
| **1.** program | **2.** begin | **3.** var | **4.** for | **5.** then | **6.** until |

1 mark each (the list of page 18). Total, num1, Sum, avg, Name and Marks are identifiers.

## E Find the errors

5 marks

| Line | Correct statement | Why |
|---|---|---|
| 1 | `program addNum (input,output);` | no space in an identifier |
| 3 | `num1,num2,total : integer;` | a semicolon ends the statement |
| 5 | `Writeln('Enter First Number');` | a string goes in single quotes |
| 9 | `total := num1 + num2;` | := is the assignment operator |
| 11 | `End.` | the program ends with End. (a full stop) |

1 mark for each error found and corrected. Pascal is not case sensitive, so Total and total are the same identifier - not an error.

MARKING SCHEME

# Unit 1 Part 1 Test

Total 40 marks

## 1 Choose the correct answer

8 marks (1 each)

| | |
|---|---|
| **1.** (c) input | **2.** (b) an algorithm |
| **3.** (c) repetition | **4.** (c) diamond |
| **5.** (c) DISPLAY | **6.** (c) Num1 |
| **7.** (d) String | **8.** (a) 2 |

## 2 Fill in the blanks

8 marks (1 each)

| | | | |
|---|---|---|---|
| **1.** solution space | **2.** sequence | **3.** parallelogram | **4.** pseudo code |
| **5.** const | **6.** := | **7.** Boolean | **8.** lost |

## 3 Work out the results

**8 marks (1 each)**

| | | | |
|---|---|---|---|
| **1.** 7 + 3 * 2 = **13** | **2.** 25 DIV 4 = **6** | **3.** 25 MOD 4 = **1** | **4.** 9 / 2 = **4.5** |
| **5.** 18 / 3 * 2 = **12** | **6.** (5 > 3) AND (2 > 4) = **False** | **7.** (5 > 3) OR (2 > 4) = **True** | **8.** NOT (6 MOD 2 = 0) = **False** |

## 4 Answer these

**16 marks**

**1. Identify the input, the process and the output of "finding the area of a circle". (3 marks)**

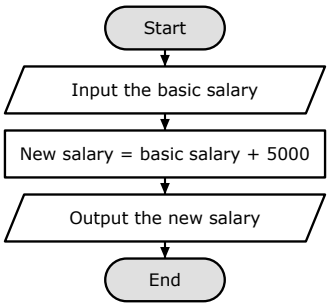
Input: the radius of the circle (1). Process: calculate the area: Area = 22/7 × radius × radius (1). Output: the area of the circle (1).

**2. Name the three control structures. Give one example of each. (3 marks)**

Sequence (climbing a staircase step by step), selection (35 or more is a pass, if not a fail) and repetition (marking the register till the last name). 1 mark each, with an example.

**3. Draw a flowchart: input the basic salary of an employee, add Rs. 5000 and output the new salary. (4 marks)**

Start; Input the basic salary (parallelogram); New salary = basic salary + 5000 (rectangle); Output the new salary (parallelogram); End. 1 mark for Start and End, 1 for each of the other three symbols.




**4. Look at the program. What is its output? What does := do? (3 marks)**

The output is 3 2 (2 marks: 17 DIV 5 = 3, 17 MOD 5 = 2). := is the assignment operator: it assigns the value on the right to the variable on the left; the previous value of a is lost (1).

```
var a, b : integer;
begin
    a := 17;
    b := a MOD 5;
    a := a DIV 5;
    writeln(a, ' ', b);
end.
```

**5. Write two differences between a variable and a constant. (3 marks)**

A variable can take different values while the program is executed; the value of a constant remains unchanged (2). A variable is declared with var; a constant is declared with const (1).

### TEACHING TIPS

- An array declared [0..4] has 5 elements. Ask often: how many elements? What is the last index?
- For rep_sel and ictMarks, add e_count := 0; o_count := 0; and tot := 0; - good habits for every counter.
- Procedure or function? A function gives an answer back (like a calculator key); a procedure just does a job.
- Revise the word lists with a quick game: you read a meaning, the class says the word.

ANSWERS - PART 2

# Chapter 4

Selection and Repetition

## A What is the output?

8 marks

| Program | Input | The last line of the output |
|---|---|---|
| positiveNo | 25 | **Positive Number** |
| positiveNo | -4 | **Enter Number** |
| LargeNo | 18 27 | **Large Number is 27** |
| GradeForMarks | 68 | **Grade = B** |
| GradeForMarks | 34 | **Grade = F** |
| FindGrade | 42 | **Grade is S** |
| FindGrade | 75 | **Grade is A** |
| FindGrade | 120 | **Invalid Marks** |

1 mark each. Every program first prints its prompt (Enter Number ...). With -4, N > 0 is false, so positiveNo prints nothing more. 120 is in no range of the CASE, so Else prints Invalid Marks and the last IF prints nothing.

## B Trace the loops

6 marks

**OUTPUT**

```
ICT
ICT
ICT
```

**OUTPUT**

```
5
10
15
20
```

**OUTPUT**

```
10
7
4
1
```

2 marks each. Script 1 runs until count > 2 (count 1, 2, 3): ICT three times. Script 3: 10, 7, 4, 1 - then n is -2 and the loop stops.

## C Complete the program

6 marks

| | | | | | |
|---|---|---|---|---|---|
| **1.** to | **2.** begin | **3.** mod | **4.** then | **5.** else | **6.** end |

```
program evenOdd(input,output);
var num, count : integer;
begin
    for count := 1 to 5 do
    begin
        writeln('Enter Number');
        read(num);
        if num mod 2 = 0 then
            writeln('Even number')
        else
            writeln('Odd number');
    end;
end.
```

1 mark each. The block of the FOR loop is between begin and end;. There is no ; before else.

## D Find the errors

6 marks

| Line | Correct statement | Why |
|---|---|---|
| 1 | `program LargeNo(input,output);` | no space in an identifier |
| 2 | `Var N1,N2,Large: integer;` | the statement ends with ; |
| 6 | `If N1 > N2 then` | IF needs then |
| 7 | `Large := N1` | := assigns a value |
| 10 | `Writeln('Large Number is ', Large);` | strings in single quotes |
| 11 | `End.` | End. with a full stop |

1 mark for each error found and corrected (the program LargeNo, page 27).

ANSWERS - PART 2

# Chapter 5

Nested Structures and Arrays

## A Fill the array

7 marks

| marks[1] | marks[2] | marks[3] | marks[4] | marks[5] |
|---|---|---|---|---|
| 20 | 30 | 15 | 55 | 40 |

1. How many elements does the array marks have? **5**
2. What is the index of the first element? **1**

1 mark for each element: marks[2] = 15 * 2 = 30; marks[4] = 40 + 15 = 55. 1 mark for each question.

## B What does orderNos print?

6 marks

OUTPUT

```
Select Ascending(A) or Descending(D)
Descending Order
6
5
4
3
2
1
```

OUTPUT

```
Select Ascending(A) or Descending(D)
Ascending Order
1
2
3
4
5
6
```

3 marks each: 1 for the prompt and the heading, 2 for the six numbers in the right order. A small a is not equal to 'A', so nothing more is printed.

## C What does it print?

6 marks

OUTPUT
```
36
```

OUTPUT
```
105
```

OUTPUT
```
6
```

OUTPUT
```
75
36
81
```

OUTPUT
```
60
81
36
```

OUTPUT
```
72
```

1 mark each (num = 45, 75, 36, 81, 60).

## D Complete the program

5 marks

| | | | | |
|---|---|---|---|---|
| **1.** 0..9 | **2.** 0 | **3.** 0 | **4.** < | **5.** min |

```
program lowMarks(input,output);
var marks : array[0..9] of integer;
    i, min : integer;
begin
    for i := 0 to 9 do
        read(marks[i]);
    min := marks[0];
    for i := 1 to 9 do
        if marks[i] < min then min := marks[i];
    writeln('Lowest mark = ', min);
end.
```

1 mark each. min starts as the first element; every element that is smaller takes its place (like max in ictMarks, page 38).

ANSWERS - PART 2

# Chapter 6

Sub-programs and Programming Languages

## A Procedure or function?

6 marks

| | | | | | |
|---|---|---|---|---|---|
| **1. Function** | **2. Procedure** | **3. Procedure** | **4. Function** | **5. Function** | **6. Procedure** |

1 mark each (1.10.1, pages 39 - 40).

## B Which language level?

8 marks

| | | | |
|---|---|---|---|
| **1. Machine** | **2. Assembly** | **3. High-level** | **4. Machine** |
| **5. High-level** | **6. High-level** | **7. Machine** | **8. Assembly** |

1 mark each (1.11.2 and 1.11.3, pages 43 - 44).

## C Assembler, interpreter or compiler?

6 marks

| | | |
|---|---|---|
| **1. Assembler** | **2. Interpreter** | **3. Compiler** |
| **4. Interpreter** | **5. Compiler** | **6. Interpreter** |

1 mark each (1.11.5, pages 49 - 50).

### TEACHING TIP

- Show procedure_circle and function_circle side by side on the projector: same output, two methods.

MARKING SCHEME

# Unit 1 Part 2 Test

Total 40 marks

## 1 Choose the correct answer

8 marks (1 each)

| | |
|---|---|
| **1.** (b) CASE | **2.** (b) 6 |
| **3.** (a) at the beginning | **4.** (b) 5 |
| **5.** (b) function | **6.** (a) machine |
| **7.** (a) PHP | **8.** (b) compiler |

## 2 Fill in the blanks

8 marks (1 each)

| | | | |
|---|---|---|---|
| **1.** nested | **2.** downto | **3.** true | **4.** infinite |
| **5.** 4 | **6.** before | **7.** low | **8.** compiler |

## 3 What is the output?

8 marks (2 each)

**OUTPUT**
```
Odd
```

**OUTPUT**
```
6
4
2
```

**OUTPUT**
```
27
```

**OUTPUT**
```
13
```

Part 3: k is 1, 3, 9, 27 - the loop stops when k is 27 (27 < 20 is false).

## 4 Answer these

16 marks

**1. Write a FOR loop that prints the numbers from 10 down to 1. (2 marks)**

for count := 10 downto 1 do writeln(count); (1 mark for the FOR ... DOWNTO line, 1 for writeln).

**2. Rewrite this IF statement as a CASE statement for the marks 0..100 (0..49: grade 'S', 50..100: grade 'P'; any other mark: Invalid Marks). (4 marks)**

Case Marks of 0..49 : Grade := 'S'; 50..100 : Grade := 'P'; Else Writeln('Invalid Marks'); End; (1 Case ... of, 1 each range line, 1 Else ... End;).

```
Case Marks of
    0..49 : Grade := 'S';
    50..100 : Grade := 'P';
Else
    Writeln('Invalid Marks');
End;
```

**3. Declare an array for 5 whole numbers and write a FOR loop that reads 5 numbers into it. (4 marks)**

var num : array[0..4] of integer; (2 marks) for i := 0 to 4 do read(num[i]); (2 marks). Accept [1..5] with for i := 1 to 5.

**4. Write one difference between a procedure and a function. (2 marks)**

A function returns an output back to the main program; a procedure does not (2).

**5. Write two differences between an interpreter and a compiler. (4 marks)**

An interpreter translates and executes one statement at a time, each time the program runs; a compiler translates the whole program once before it runs (2). With a syntax error an interpreter runs till the error; a compiler does not translate the program and highlights the errors (2). After compiling, the program can be executed any number of times.

### (i) WHAT STUDENTS USUALLY GET WRONG

- They mix up input and process: the input is what you start with; the process is what you do with it.
- They use the parallelogram for a process. Rectangle = process; parallelogram = input or output.
- They write = for an assignment in Pascal. In Pascal it is := (= compares).
- They put a semicolon before ELSE, or forget THEN after the condition of an IF.
- They count FOR X := 5 TO 10 as 5 repetitions. It is 6 (10 - 5 + 1).
- They think REPEAT ... UNTIL stops when the condition is false. It stops when it becomes TRUE.
- They say array[0..4] has 4 elements. It has 5: num[0] to num[4].
- They mix up the interpreter (one statement at a time, every run) and the compiler (the whole program once).

### (!) TEXTBOOK MISPRINTS CORRECTED IN THE WORKBOOKS

- p. 16 - 17: inside the loop of the smallest-number example, "Output the number N" is an INPUT.
- p. 18: "ate" and the second "to" at the end of the reserved words are not Pascal words.
- p. 24: total: =num1 + num 2; is total := num1 + num2; p. 28: "Grade = " needs single quotes.
- p. 28: the CASE flowchart gives "W" for 35 - 49; the program (p. 29) gives 'S'.
- p. 24 E.g. 5: 4 >= 4 AND NOT(7 > 9) needs brackets in Pascal (AND comes before >=): (4 >= 4) AND NOT (7 > 9).
- p. 30: print Reverse has a space - printReverse. For DOWNTO, Value_1 must be LARGER than Value_2 (the textbook says smaller). p. 32: count = 0; is count := 0;
- p. 34: Assending / Desending are spelt Ascending / Descending. p. 35: writeln((o_count has an extra (.
- p. 37: num[2] := 36,num[4] := 60; needs a ; . p. 38: writeln(num[1], num[4]) prints 75 and 60 (not 36).
- p. 40: V = sub2(value); is V := sub2(value); p. 41: Circumferance is spelt Circumference.
- rep_sel and ictMarks never set e_count, o_count and tot to 0: Free Pascal starts them at 0, but add the statements in class.

## HOMEWORK AND EXTRA PRACTICE

- Part 1: draw a flowchart and write the pseudo code for "the first 12 multiples of 7" (Activity, page 12).
- Part 1: list five valid and five invalid identifiers with the rule each one breaks.
- Part 2: write a program that prints the multiples of 5 from 5 to 50 with FOR, WHILE and REPEAT.
- Part 2: write a function that returns the area of a rectangle and call it from the main program.
- Past O/L papers: the questions on algorithms, flowcharts, Pascal and the language translators.

## TEACHING TIPS

- Talk to a computer in binary for one minute: students feel why high-level languages were invented.
- Interpreter vs compiler: a translator who translates sentence by sentence vs one who translates the whole book first.
- O/L papers ask about the language translators and the control structures every year. Revise them often.
- Mark the part tests together the next day. Students learn most from the questions they got wrong.
- Read the writing-space notes aloud slowly, or write them on the board; check the students' notes.
- Keep the textbook open at the right page in every practical. The page numbers are in the practical titles.
- Pairs work well in the lab: one types, one reads the program aloud. Swap after each program.
- Students who finish early can change a program: grades for 0..100 with CASE, or the circle with the radius 14.
- Turn every flowchart into a pseudo code on the board, symbol by symbol: a diamond with a loop is WHILE.
- Let students write the pseudo code first and the Pascal program second. The pseudo code is the plan.
- Use the school timetable as a sequence, the school canteen menu as a selection, the morning assembly as a repetition.
- Ask: which data type for a telephone number, a grade, a price, a name, Yes or No? Discuss the answers.
- Show a program with good and bad identifiers (x1, total_marks). Which one is easier to read?
- In the lab, check that every program begins with program ... ; and ends with end. (a full stop).
- For arrays, use the class register: the index is the register number, the element is the student.
- Before the part test, let pairs ask each other the questions of the revision page.